Machine Learning

Challenge 2026

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The challenge

The taks consists in a multiclass classification problem.

Objective

Estimate the recommended fertilizer type given crop and soil types, temperature, humidity, etc.

id	Temparature	Humidity	Moisture	Soil Type	Crop Type	Nitrogen	Potassium	Phosphorous	Fertilizer Name
0	37	70	36	Clayey	Sugarcane	36	4	5	28-28
1	27	69	65	Sandy	Millets	30	6	18	28-28
2	29	63	32	Sandy	Millets	24	12	16	17-17-17
3	35	62	54	Sandy	Barley	39	12	4	10-26-26
4	35	58	43	Red	Paddy	37	2	16	DAP

Figure: Dataset extract

Reminder: you may choose either this challenge (probably more suited to ESP students) or the Beacon.io challenge (probably more suited to CNN students).

The data

- Training data: 750k samples. 7 possible labels.

id	Temparature	Humidity	Moisture	Soil Type	Crop Type	Nitrogen	Potassium	Phosphorous	Fertilizer Name
0	37	70	36	Clayey	Sugarcane	36	4	5	28-28
1	27	69	65	Sandy	Millets	30	6	18	28-28
2	29	63	32	Sandy	Millets	24	12	16	17-17-17
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4	35	58	43	Red	Paddy	37	2	16	DAP

- Test data: 250k *unlabeled* samples. Labels to be predicted.

id	Temparature	Humidity	Moisture	Soil Type	Crop Type	Nitrogen	Potassium	Phosphorous
750000	31	70	52	Sandy	Wheat	34	11	24
750001	27	62	45	Red	Sugarcane	30	14	15
750002	28	72	28	Clayey	Ground Nuts	14	15	4
750003	37	53	57	Black	Ground Nuts	18	17	36
750004	31	55	32	Red	Pulses	13	19	14

(a) Example test set

id	Fertilizer Name
750000	14-35-14 28-28 DAP
750001	17-17-17 Urea 28-28
750002	20-20 14-35-14 17-17-17
750003	10-26-26 17-17-17 14-35-14
750004	20-20 10-26-26 28-28

(b) Corresponding preds.

Submission procedure

- Train and test datasets are available here: `https://www.kaggle.com/competitions/playground-series-s5e6/overview`
- To receive a score, you need to submit a file containing the predictions of your model for the test dataset.
- (You will probably need to create an account for your team)

id	Fertilizer Name
750000	14-35-14 28-28 DAP
750001	17-17-17 Urea 28-28
750002	20-20 14-35-14 17-17-17
750003	10-26-26 17-17-17 14-35-14
750004	20-20 10-26-26 28-28

Figure: Example submission of 3 estimated labels

Grading

The project grade will consist of:

- ∼50% score on the leaderboard.
  - Please attach a screenshot of your score to your report.
- ∼50% report.
  - Cf. next slides.

Exact grading scheme may be slightly adjusted. This is usually to your advantage.

Score

- Metrics used for scoring: same as Kaggle challenge, i.e. mean average precision @3:

$$\text{MAP@}K = \frac{1}{N}\sum_{n=1}^{N}\sum_{k=1}^{K} P(k)1_{\text{relevant}}(k)$$

- i.e. your top 3 predicted labels should ideally contain the correct label.
  - cf. challenge rules on Kaggle for more detailed explanations.
- The final grading scheme will be determined after the challenge is over.
  - But in general, the higher the score, the higher the grade.
- Exceptions may apply (cf. later slide).
- You (probably) don't need to rank in the top of the public leaderboard to get a good grade.

Expected deliverables

To be sent by email:

- Your report (cf. next slide)
- The code used for the final submission
  - Ideally, alongside the report, not *in* the report
  - In a .zip file, as a .ipynb notebook, or through a link to a git repository
  - Warning: .py files are blocked by the email system. If you need to include .py files, please include them in a .zip file.
- Your final Kaggle submission file, as a .csv file.
- A screenshot of your score on the Kaggle test set.

Strict rule

Anything submitted (report, code, etc.) *must* be your own. Anything that is not your own (AI generated content, copy-pasted content, etc.) *will not be graded*.

Report

Report

Your report should resemble a scientific paper

It should contain:

- Detailed explanation of the methodology used for final submission
  - Preprocessing steps, feature engineering, choice of model, choice of hyperparameters, etc.
- Any observation and/or exploration about the dataset which you think is relevant to your choices.
- Optionally, other ideas you tried and why you discarded them (cf. next slide).
- Any other graph, figure, study etc which you consider to be of interest.

Additional rules

- You may look at other resources (e.g. blog posts about this challenge or this dataset), but you should cite them in your report if you end up using them.
- Again, anything submitted *must still* be your own.
- You may use additional data, pre-trained models, etc, but you should explicitly mention this and justify your choice.
- Score may be evaluated on a case by case basis based on this information.
- If you are unsure about the rules, please ask.

Additional information

- Candidate deadline: 10th January 2027 at midnight.
  - May be subject to change
  - You will be warned in advance if this is the case
- Groups of 2 students.
  - Some exceptions may be allowed *on request*.

- An optional lab session may be organized on request if you have questions about the challenge or the data.